```
#include <AFMotor.h>
#include <Servo.h>
Servo myservo;
int pos = 65;

AF_DCMotor motor1(1, MOTOR12_1KHZ);
AF_DCMotor motor2(2, MOTOR12_1KHZ);
AF_DCMotor motor3(3, MOTOR34_1KHZ);
AF_DCMotor motor4(4, MOTOR34_1KHZ);


void setup()
{
  Serial.begin(9600);
  myservo.attach(10);
  myservo.write(90);
  //Set initial speed of the motor & stop
  motor1.setSpeed(250);
  motor1.run(RELEASE);
  motor3.setSpeed(250);
  motor3.run(RELEASE);
  motor2.setSpeed(255);
  motor2.run(RELEASE);
}


void loop()
{




  // digitalWrite (2, LOW);



  // digitalWrite (2, HIGH);
  // delay(1000);

  // delay(1000);
  int sensorReading = analogRead (A0);
  int sensorReading2 = analogRead (A1);

  // Map sensor value between range 0-3
  int range = map (sensorReading, 0, 1024, 0, 3);

  if (range == 0) {
    Center();
```

```
  Serial.print("0");
 }
 if (range == 1 ) {
  Serial.print("1");
  Center();
 }
 if (range == 2 ) {


  Serial.print("2");
 }

 int range2 = map (sensorReading2, 0, 1024, 0, 3);

 if (range2 == 0) {
  Center();

  Serial.print("0");
 }
 if (range2 == 1 ) {
  Serial.print("1"); Center();

 }
 if (range2 == 2 ) {

  Serial.print("2");
 }


}
void Center() {


 // digitalWrite (2, LOW);
 motor1.run(FORWARD);
 motor1.setSpeed(180);
 motor3.run(FORWARD);
 motor3.setSpeed(180);
 delay(1000);

 motor3.run(RELEASE);
 motor1.run(RELEASE);

 motor2.run(FORWARD);
 motor2.setSpeed(255);
```

```
  for (pos = 50; pos <= 130; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 130; pos >= 50; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 50; pos <= 130; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 130; pos >= 50; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }

  delay(200);
   myservo.write(90);
  motor2.run(RELEASE);
}

void Right() {

   // digitalWrite (2, LOW);
  motor1.run(FORWARD);
  motor1.setSpeed(200);
  motor3.run(FORWARD);
  motor3.setSpeed(200);
  delay(1000);

  motor3.run(RELEASE);
  motor1.run(RELEASE);

  motor2.run(FORWARD);
  motor2.setSpeed(255);

  for (pos = 50; pos <= 130; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 130; pos >= 50; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
```

```
  for (pos = 50; pos <= 130; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 130; pos >= 50; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }

  delay(200);
   myservo.write(90);
  motor2.run(RELEASE);

}

void Left() {

  motor2.run(FORWARD);
  motor2.setSpeed(255);
  // digitalWrite (2, LOW);

  motor3.run(FORWARD);
  motor3.setSpeed(120);
  motor1.run(FORWARD);
  motor1.setSpeed(150);
  delay(700);

  motor1.setSpeed(200);
  motor1.run(RELEASE);
  motor3.setSpeed(200);
  motor3.run(RELEASE);
  for (pos = 60; pos <= 120; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 120; pos >= 60; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  motor2.setSpeed(255);
  motor2.run(RELEASE);
  delay(2000);



}
```

```
void centerservo() {
  for (pos = 65; pos <= 150; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 150; pos >= 65; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);
  }
  for (pos = 65; pos <= 150; pos += 1) { // goes from 0 degrees to 180 degrees
    // in steps of 1 degree
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);                       // waits 15 ms for the servo to reach the position
  }
  for (pos = 150; pos >= 65; pos -= 1) { // goes from 180 degrees to 0 degrees
    myservo.write(pos);              // tell servo to go to position in variable 'pos'
    delay(15);
  }

}
```